

The Consequences of Ideas Being Harder to Find: Evidence from US Patenting

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Abstract

As ideas become harder to find, firms shift R&D from frontier-pushing innovation toward activities that protect incumbent market position. I embed this mechanism in a Klette-Kortum growth model where multi-product firms allocate research labor between productive R&D, which improves quality and drives growth, and defensive R&D, which accumulates displacement barriers without improving efficiency. Productive R&D faces an increasing fishing-out penalty while defensive R&D does not. The resulting shift toward defense generates rising markups, slowing productivity growth, and declining business dynamism. I present reduced-form evidence of the model predictions. Using firm-level data matched to 2.9 million patents and instrumenting with patent examiner leniency, I find that increases in defensive R&D cause markups to rise with a 4–6 year lag and TFP to decline significantly from horizon 4 onward. Measures of business dynamism show the same result. Firm-level descriptive facts support the conclusion. Defensive R&D is associated with higher revenue productivity but not with higher costs, whereas productive R&D is.

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1 Introduction

Several trends characterize the U.S. economy over the past decades, while the following ones are relevant for this paper: Markups have risen, with the sales-weighted mean climbing from around 1.05 in 1985 to 1.25 in 2020 (De Loecker et al., 2020). Productivity has seen a decline, with TFP growth falling from roughly 1.5 percent per year in the 1960s to below 0.5 percent after 2005, despite a doubling of R&D spending as a share of GDP (Bloom et al., 2020). Business dynamism has declined, with the entry rate falling from over 13% in 1980 to under 9% by 2020 (Decker et al., 2020). And the composition of R&D has shifted toward incremental activities, concentrated among large firms (Akcigit and Kerr, 2018; Kelly et al., 2021).

This paper proposes a unified explanation. I build a Klette-Kortum growth model in which multi-product firms allocate research effort between productive R&D, which improves quality and hence drives growth, and defensive R&D, which sole purpose is to accumulate displacement barriers. The key asymmetry is that productive R&D becomes progressively harder as the frontier advances, building on the logic of "ideas getting harder to find" from Bloom et al. (2020), whereas defensive R&D does not. This tilts the relative return toward defense, generating an endogenous shift that can simultaneously produces the relevant trends. My model yields four testable predictions. First, the composition of R&D shifts toward defense while productivity growth falls. Second, markups rise (with a lag) as barriers accumulate gradually. Third, business dynamism declines as higher barriers reduce displacement. Fourth, large firms allocate more to defense because the shielding value of barriers is increasing in the incumbent's profit flow. I provide reduced-form evidence by estimating local projections instrumented with quasi-random patent examiner leniency, as established by Sampat and Williams (2019). Exogenous increases in defensive R&D (as proxied by non-breakthrough patents) cause markups to rise, peaking at horizons 5–6, and TFP to decline significantly from horizon 4 onward. The zero-on-impact pattern matches the model's accumulation mechanism and weakens reverse causality concerns. Using Census Business Dynamics Statistics, I show that all six dynamism measures decline uniformly. At the mechanism level, firm-level regressions show that defensive R&D raises revenue productivity with zero effect on cost-based productivity.

Three key contributions follow: First, it shows that the R&D–growth relationship depends critically on the composition of R&D. When firms redirect effort from productive to defensive activities, aggregate spending can rise while growth falls. This parallels the findings of De Ridder (2024) that the distribution of R&D across firms matters, but operates through the within-firm margin. Second, the model distinguishes between displacement barriers reflecting genuine efficiency and barriers representing socially wasteful investment, with direct policy implications. Third, the fishing-out asymmetry provides an

endogenous mechanism for the declining innovation quality of laggard firms documented by Olmstead-Rumsey (2022).

Related Literature: This paper connects to two strands of the literature. On the theoretical side, the model builds on the idea of creative destruction. On the empirical side, it relates to papers documenting, either individually or jointly, the trends discussed above as well as other trends.¹

With regard to the theoretical literature, the starting point for this paper is the creative-destruction framework of Aghion and Howitt (1992), where long-run growth is driven by innovations that displace incumbents. Klette and Kortum (2004) extend this logic to a multi-product firm environment capable of matching key firm-level facts such as skewed firm-size distributions and approximately size-independent R&D intensity. More recent work studies how the composition of innovation shapes firm dynamics and market structure. Akcigit and Kerr (2018) distinguish between internal and external innovation, showing that large firms concentrate on improvements within existing product lines while smaller firms drive more disruptive innovation. Their results motivate the distinction between productive and defensive R&D in my framework, although the mechanism differs: both forms of innovation improve quality in their model, whereas defensive R&D in mine does not. Peters (2020) studies how the interaction between incumbent innovation and creative destruction shapes the endogenous distribution of markups.

A large empirical body further documents the broad trends motivating this paper. De Loecker et al. (2020) show that U.S. markups have risen substantially since the 1980s, particularly in the upper tail of the distribution, while Autor et al. (2020) document the rise of superstar firms. At the same time, Decker et al. (2014) show that business dynamism, measured through entry, exit, and job reallocation, has steadily declined. On the innovation side, Bloom et al. (2020) document falling research productivity, arguing that ideas are becoming progressively harder to find. More recent literature attempts to explain these trends jointly. Akcigit and Ates (2023) study several candidate mechanisms behind declining dynamism and rising concentration in a quantitative innovation model, highlighting reduced knowledge diffusion as the dominant force. Importantly for this paper, their “ideas being harder to find” mechanism counterfactually lowers concentration. Olmstead-Rumsey (2022) shows that declining innovation quality among laggard firms can account for a large share of both the rise in concentration and the slowdown in productivity growth. Baqaee and Farhi (2020) further show that rising markups amplify misallocation in general equilibrium. The paper is most closely related to De Ridder (2024). In his model, firms differ in intangible efficiency by endogenously drawing an efficiency parameter at birth. This generates displacement barriers because challengers must overcome both quality differences and incumbents’ cost advantages. In contrast, the

¹For a good discussion of further trends, see Akcigit and Ates (2023).

barriers in my framework arise endogenously through firms' allocation of R&D toward defensive activities. This distinction is important for welfare, as having an advantage in the market due to efficiency in deploying intangible capital is not necessarily a socially wasteful thing.

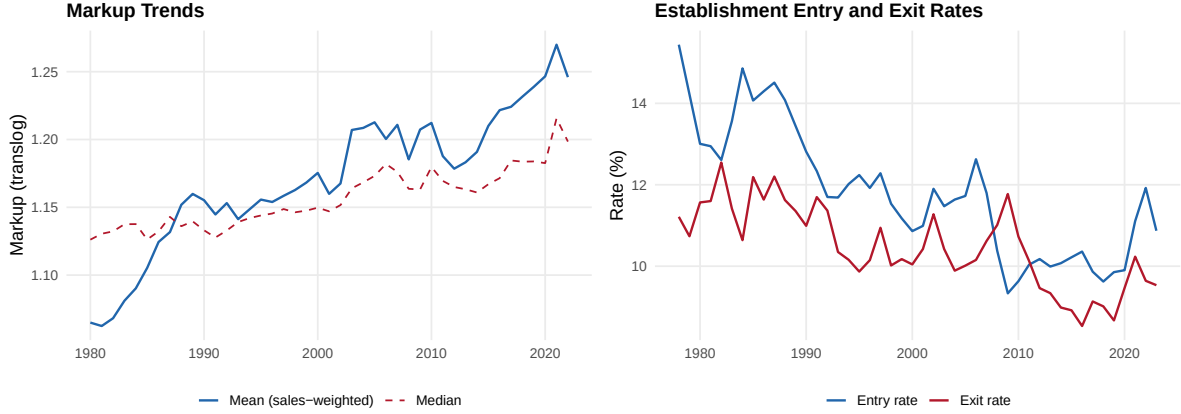


Figure 1: Markups and Business Dynamism Over Time

Notes. Left panel shows sales-weighted mean and median translog markups for all Compustat firms excluding financials and utilities, 1980–2022. Right panel shows economy-wide establishment entry and exit rates from Census BDS, 1978–2023.

Motivating Evidence: Figure 1 (left) plots sales-weighted mean and median markups for listed U.S. firms between 1980 and 2022. The mean rose from around 1.05 to roughly 1.25, with the increase concentrated in the upper tail of the firm size distribution, as discussed by De Loecker et al. (2020). Although recent studies indicate that the magnitude of these estimates is biased, revenue data can still be used to reliably determine their trend and distribution. (De Ridder et al., 2026). The right panel plots establishment entry and exit rates over the same time span, taken from the BDS (Decker et al., 2020). Entry fell from over 13 percent in 1980 to under 9 percent by 2020, with exit declining in parallel. Other measures of business dynamism, such as job reallocation rates or average firm age show the same

Figure 2 documents R&D composition. The left panel shows the patent-weighted non-breakthrough share (stable around 85–90%). The right panel shows R&D intensity (XRD/sales), which has risen steadily. A stable non-breakthrough share combined with rising total R&D means that the absolute level of defensive R&D has grown while productive R&D has stagnated.

2 Model Setup

Preferences and Market Structure. Time is continuous. A representative household maximizes $U = \int_0^\infty e^{-\rho t} \ln C_t dt$ with discount rate $\rho > 0$. Labor is supplied inelastically,

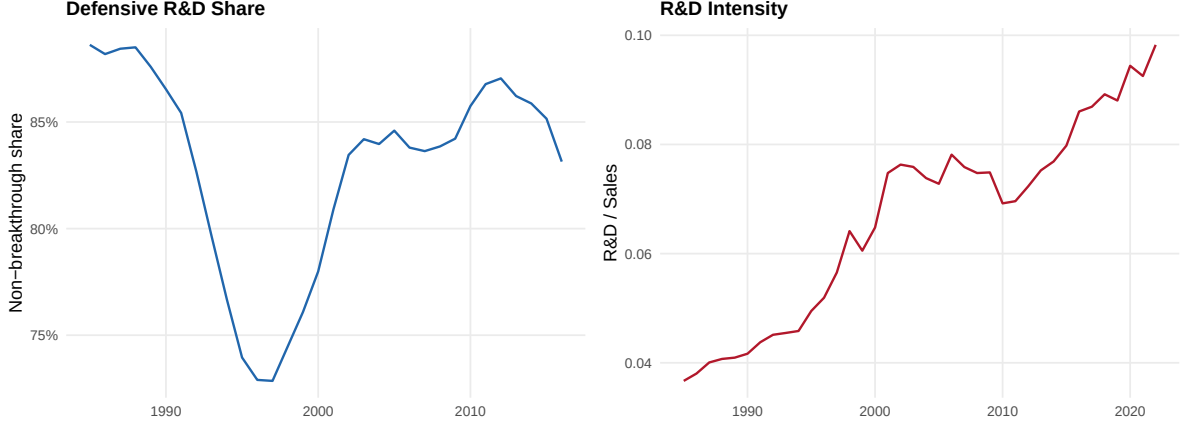


Figure 2: R&D Composition and Intensity

Notes. Left panel shows the patent-weighted non-breakthrough share among Compustat firms with KPST scores, 1985–2022, using the text-based importance measure of Kelly et al. (2021). Right panel shows industry-weighted average R&D intensity (XRD/sales) across Compustat firms with patents, 1985–2022.

with one unit to production and $\bar{L}$ units to research. A unit continuum of intermediate goods $j \in [0, 1]$ is aggregated via Cobb-Douglas technology

$$Y = \exp \left(\int_0^1 \ln \left(\sum_{i \in \mathcal{I}_j} q_{ij} y_{ij} \right) dj \right), \quad (1)$$

where $\mathcal{I}_j$ is the set of firms producing good j , q_{ij} is firm i 's quality, and y_{ij} is quantity. Firms compete à la Bertrand. The Cobb-Douglas aggregator implies unit-elastic demand and equal expenditure on each good.

Firms, Products, and Displacement Barriers. Following Klette and Kortum (2004), each firm owns a product portfolio. Production is linear in labor, $y_j = q_j \ell_j$, giving marginal cost w/q_j . Under Bertrand limit pricing, the leader sets $p_j = w/q_j^{(2)}$, just undercutting the runner-up's marginal cost. Defining the quality gap $\lambda_j \equiv q_j/q_j^{(2)}$, the markup over own marginal cost is $\mu_j = p_j/(w/q_j) = \lambda_j$, and flow profit on line j is $\pi_j = (1 - \lambda_j^{-1})Y$.

Each product also carries a displacement barrier $b_j \geq 1$, accumulated through defensive R&D. The barrier does not enter the production function: it neither lowers costs nor improves quality. A potential innovator with quality step λ displaces the incumbent on product j only if $\lambda > b_j$. When displacement fails, the innovation is lost.

Productive R&D. A firm with n products choosing Poisson innovation rate x per product employs

$$\ell^P(x, n) = n \cdot a x^\gamma (1 + \phi Q)^{\gamma-1} \quad (2)$$

researchers, where $\gamma > 1$ governs curvature, $Q = \int_0^1 q_j dj$ is the aggregate quality index, and $\phi > 0$ is the fishing-out parameter. The $(1 + \phi Q)^{\gamma-1}$ term raises the cost of productive innovation as the frontier advances. A successful innovation gives quality $q_{ij} = q_j^{(1)} \cdot \lambda_{ij}$ where $\lambda_{ij} \sim H(\lambda)$ on $[1, \bar{\lambda}]$, and the innovator displaces the incumbent only if $\lambda_{ij} > b_j$. The linearity of ℓ^P in n ensures that per-product R&D intensity is independent of firm size, consistent with Klette and Kortum (2004).

Defensive R&D. Each product's barrier is built independently. A firm investing d_j units of research labor on product j obtains:

$$b_j = (1 + c \cdot d_j)^{1/\alpha}, \quad c > 0, \alpha > 1 \quad (3)$$

The function $b(d)$ is increasing and concave. Defensive R&D faces no fishing-out penalty. This asymmetry is the engine of the model. Total defensive labor for a firm with n products is $\ell^D = \sum_{j=1}^n d_j$. Concavity implies the firm spreads defense across its product lines. Optimal defense depends on the line's quality gap through the profit flow, $d_j^* = d^*(\lambda_j)$ with $dd^*/d\lambda > 0$ (Prediction 4), so $\ell^D = \sum_{j=1}^n d^*(\lambda_j)$. On the BGP the cross-line distribution of (λ_j, d_j^*) is stationary. For the aggregate displacement and growth equations I summarize this cross-section by the distribution-weighted barrier $b^* \equiv \mathbb{E}_\lambda[b(d^*(\lambda))]$.

Optimal Innovation and Defense. The net profit flow from product j after deducting defensive R&D cost is

$$\pi_j^{\text{net}} = (1 - \lambda_j^{-1})Y - w_R d_j \quad (4)$$

where w_R is the research wage. Denote the aggregate innovation flow by $\iota = \sum_n M_n \cdot n \cdot x_n + e$, where the $n \cdot x_n$ term reflects that a firm with n products, each innovating at rate x_n , generates $n \cdot x_n$ total innovations landing on random product lines. The product faces destruction at rate $\tau(b_j) = \iota \cdot \bar{H}(b_j)$, where $\bar{H}(b) \equiv \Pr(\lambda > b)$ is the probability a random innovation step exceeds the barrier. I write the value function in stationary units normalized by output Y , so that flow profits and the research wage are expressed as shares of Y , with $\tilde{v} \equiv v/Y$ and $\tilde{w}_R \equiv w_R/Y$. The flow payoff is then stationary and the appropriate discount rate is $r - g = \rho$ plus the displacement hazard. Its present value on the BGP is

$$\tilde{v}(\lambda_j, d_j) = \frac{(1 - \lambda_j^{-1}) - \tilde{w}_R d_j}{\rho + \tau(b(d_j))} \quad (5)$$

The firm chooses d_j to maximize $\tilde{v}(\lambda_j, d_j)$. Differentiating (5) and setting $\partial \tilde{v} / \partial d_j = 0$ yields

$$\tilde{w}_R(\rho + \tau(b_j)) = [(1 - \lambda_j^{-1}) - \tilde{w}_R d_j^*] \cdot \iota \cdot h(b_j^*) \cdot b'(d_j^*) \quad (6)$$

where $h(b) = -\bar{H}'(b)$ is the density of the step-size distribution evaluated at the barrier. The left side is the marginal cost: one more unit of defensive labor reduces the flow payoff

by $\tilde{w}_R$, capitalized at the effective discount rate $\rho + \tau$. The right side is the marginal benefit: a higher barrier reduces the hazard of displacement, raising the expected duration of the profit stream. Equivalently, using $\tilde{v} = [(1 - \lambda_j^{-1}) - \tilde{w}_R d_j]/(\rho + \tau)$, this simplifies to

$$\tilde{w}_R = \tilde{v}(\lambda_j, d_j^*) \cdot \iota \cdot h(b_j^*) \cdot b'(d_j^*) \quad (7)$$

Creative Destruction and Growth. The aggregate displacement rate is $\bar{\tau} = \iota \cdot \mathbb{E}_\lambda[\bar{H}(b(d^*(\lambda)))] = \iota \cdot \bar{H}(b^*)$, where $\iota = \sum_n M_n \cdot n \cdot x_n + e$ and b^* is the distribution-weighted barrier defined above. The BGP growth rate is

$$g = \bar{\tau} \cdot \mathbb{E}[\ln \lambda \mid \lambda > b^*] \quad (8)$$

Defensive R&D reduces $\bar{\tau}$ directly and truncates the step-size distribution from below. Conditional on displacement, the average quality step $\mathbb{E}[\ln \lambda \mid \lambda > b^*]$ rises, partially offsetting the decline in $\bar{\tau}$. The net effect on growth is negative when the barrier effect dominates.

Definition 1. *A balanced growth path satisfies: (i) optimality of x and d for incumbents; (ii) free entry; (iii) stationarity of the firm-size distribution; (iv) labor market clearing, $\ell^P + \ell^D + L^{prod} = 1 + \bar{L}$; and (v) the Euler equation $r = \rho + g$. Comparing BGPs as the frontier Q advances yields four predictions:*

R&D composition shifts, growth falls. As the frontier Q advances, the fishing-out penalty raises the marginal cost of productive R&D, so productive intensity $x^* \propto (1 + \phi Q)^{-1}$ declines. Defensive R&D faces no such penalty. From the defense FOC (7), the fall in productive innovation lowers the aggregate flow ι and raises the value $\tilde{v}$ of an undisplaced line, which together raise the optimal barrier d^* and hence b^* when the value effect dominates the flow effect. The defensive share of research labor therefore rises. The total quantity of research labor is fixed by the endowment, so this is a shift in composition rather than an expansion. The empirically observed rise in R&D intensity (expenditure over sales, Figure 2) is consistent with this because the relative research wage rises as productive innovation becomes scarcer, raising measured R&D expenditure relative to output even as research labor is reallocated. Growth falls because $\bar{\tau}$ depends on x^* through ι , reinforced by the rising barrier b^* truncating successful displacements.

Markups rise with a lag. Higher b^* lowers $\bar{\tau}$, allowing quality gaps to accumulate. The markup distribution adjusts only as the stock of surviving incumbents changes, producing a gradual buildup.

Dynamism declines. The displacement rate $\bar{\tau}$ falls directly. Entry falls because $\bar{H}(b^*)$ declines, reducing the probability of successful displacement.

Large firms spend more on defensive. The implicit function theorem applied to (7) gives $dd^*/d\lambda > 0$: defense is increasing in product value.

3 Empirical Analysis

Data: I combine multiple datasets, covering a time span of 1985-2022. The primary source contains data on the universe of listed US firms from Compustat (241,823 firm-years excluding financials and utilities). I match 2,899,569 patents using the Kogan et al. (2017) bridge to these listed firms. The Kelly et al. (2021) text-based breakthrough measure covers 79.5% of matched patents. The primary defensive R&D measure is the KPST non-breakthrough share. At the firm level, R&D composition is constructed by interacting R&D intensity (XRD/sales) with the breakthrough and non-breakthrough shares. For the instrument, I use PatEx, which provides examiner identifiers for 9.3 million utility applications. Leave-one-out grant rates are computed within art unit $\times$ filing year cells with at least 5 examiners, yielding 6,183 industry-year observations. The Census BDS provides six dynamism measures at the NAICS 4-digit level, collapsed to NAICS 3-digit and merged to SIC 3-digit via the official Census crosswalk. Table 3 in Appendix provides summary statistics.

Identification Strategy and Specification: I estimate the causal effect of defensive R&D on markups, TFP, and measures of dynamism using local projections (Jordà, 2005).

$$Y_{j,t+h} - Y_{j,t-1} = \beta_h \cdot \widehat{\text{DefShare}}_{j,t} + \gamma'_h \mathbf{X}_{j,t} + \alpha_j + \delta_t + \varepsilon_{j,t}, \quad h = 0, \dots, 8 \quad (9)$$

where $\widehat{\text{DefShare}}_{j,t}$ is the KPST non-breakthrough share of industry j at time t instrumented by average examiner leniency and $\{\beta_h\}$ traces the impulse response. I use LPs instead of VARs based on the arguments of Montiel Olea et al. (2025). They show that LP may lack precision but are not (or less) biased than standard VARs. Since my priority is the sign and timing of the effect, the low-bias property of LPs outweighs the variance cost. Following their recommendations, the vector of controls $\mathbf{X}_{j,t}$ contains lagged outcome, lagged defensive share, current and lagged R&D intensity, log number of firms, and log number of patents. The lag length is one and was chosen by the Akaike-Information-Criterion, though other tests yield similar lag lengths. Industry and year fixed effects are included throughout and standard errors are clustered by SIC 3-digit industry. Within the USPTO, applications are assigned to Art Units by technology class, then to individual examiners quasi-randomly conditional on Art Unit $\times$ filing year (Sampat and Williams, 2019). I compute each examiner’s leave-one-out grant rate as the instrument. A lenient examiner grants more patents at the margin, raising the non-breakthrough share since marginal patents tend to be incremental. The first-stage F -statistic is 41.2. A balance test finds that leniency does neither predict R&D intensity ($p = 0.61$) nor other covariates such as firm size as proxied by sales ($p = 0.74$). The final sample contains 3,315 industry-year observations across 177 SIC 3-digit industries, 1985–2021.

4 Results

4.1 Markups

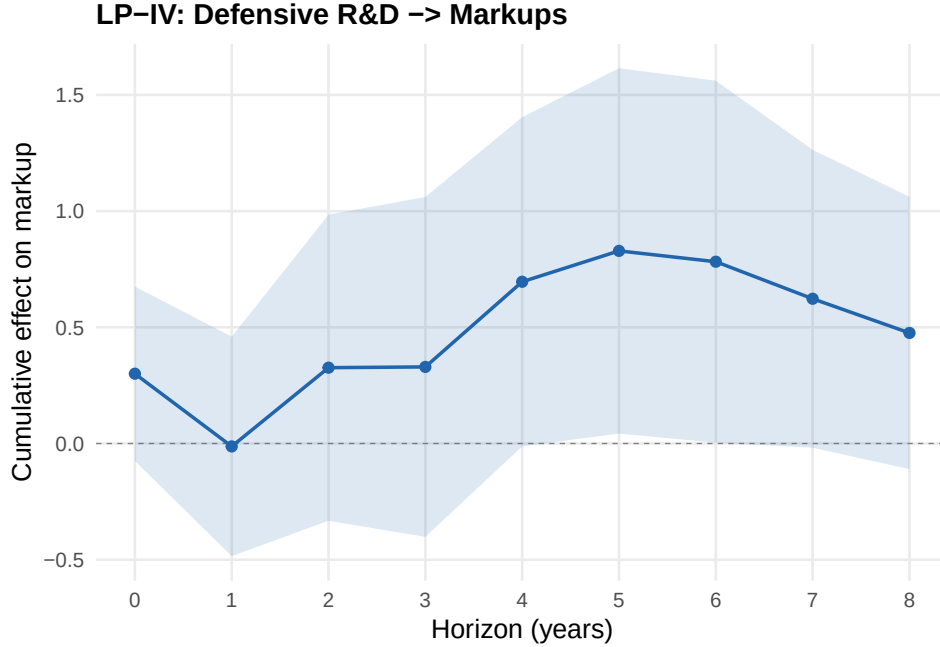


Figure 3: Impulse Response Function of Markups

Notes: Outcome is industry mean translog markup. Industry and year FE included. Standard errors clustered by SIC 3-digit. 90% CI shaded.

Figure 3 plots the LP-IV impulse response of markups to an exogenous increase in the defensive R&D share. Point estimates increase monotonically from $h = 1$ onward and peak at $h = 5$ before stabilizing through $h = 8$. The gradual buildup matches the model’s accumulation channel prediction. Additionally, the size of the effect, being between 0 and 1, also seems economically plausible.

However, some points warrant discussion. The IV identifies a local average treatment effect (LATE) among complier industries, which may differ from the average treatment effect if the defensive R&D–markup relationship is heterogeneous, i.e. we have sample bias. Given De Loecker et al. (2020) show that the rise in aggregate markups is largely driven by the right tail of the firm-size distribution, these mean estimates should be taken with caution, and the focus should be on the trend, and not the point estimates.

4.2 Total Factor Productivity

Figure 4 plots the IRF of TFP growth. TFP is roughly flat at horizons 0–2, then declines substantially from $h = 4$ onward. The timing parallels the markup response, consistent with the model as both are driven by the same mechanism (rising barriers reduce displacement, simultaneously allowing markups to accumulate and depressing growth). Since the

Solow residual partly reflects markup variation, the TFP decline could be a mechanical consequence of the markup increase. Robustness checks using alternative TFP measures that partially control for markup variation confirm the result, though full separation is infeasible with Compustat data alone.

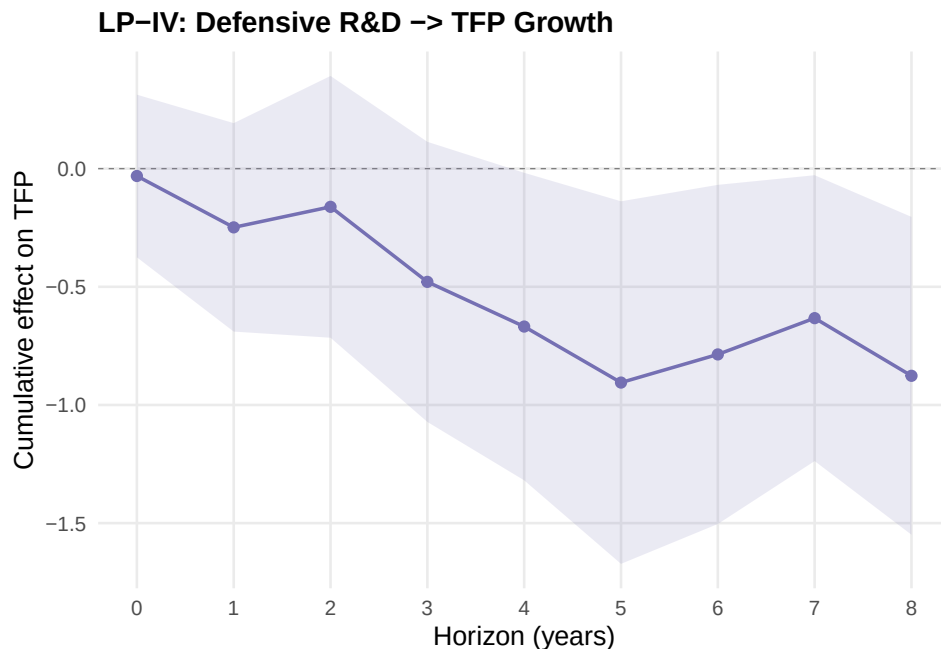


Figure 4: Impulse Response Function of TFP

Notes. TFP is the Solow residual from a production function estimated by SIC 2-digit and filtered using an HP-filter with $\lambda = 100$ (see Appendix B). Same LP-IV specification as Figure 3. 90% CI shaded.

4.3 Business Dynamism

Figure 5 plots the IRF for establishment entry and exit rates. Both decline, with the 90% CI excluding zero from horizon 4 for entry and horizon 5 for exit. Figure 6 shows that gross job reallocation declines substantially at longer horizons. Additional BDS measures (job creation from births, destruction from deaths, firm death rate) follow the same pattern.

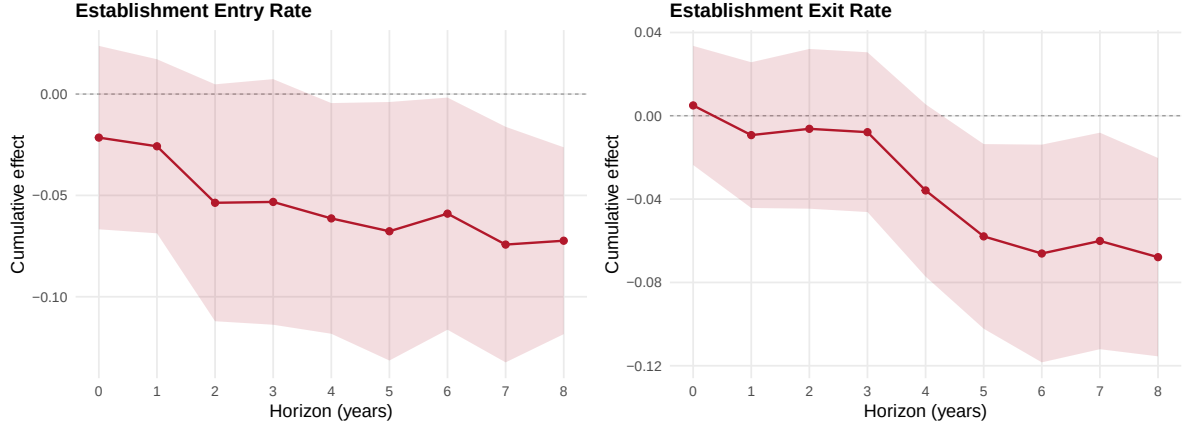


Figure 5: LP-IV. Defensive R&D → Establishment Entry and Exit Rates

Notes. Each panel shows the LP-IV impulse response of a BDS measure to the non-breakthrough share. Same specification as Figure 3. 90% CI shaded, SE clustered by SIC3 Industry.

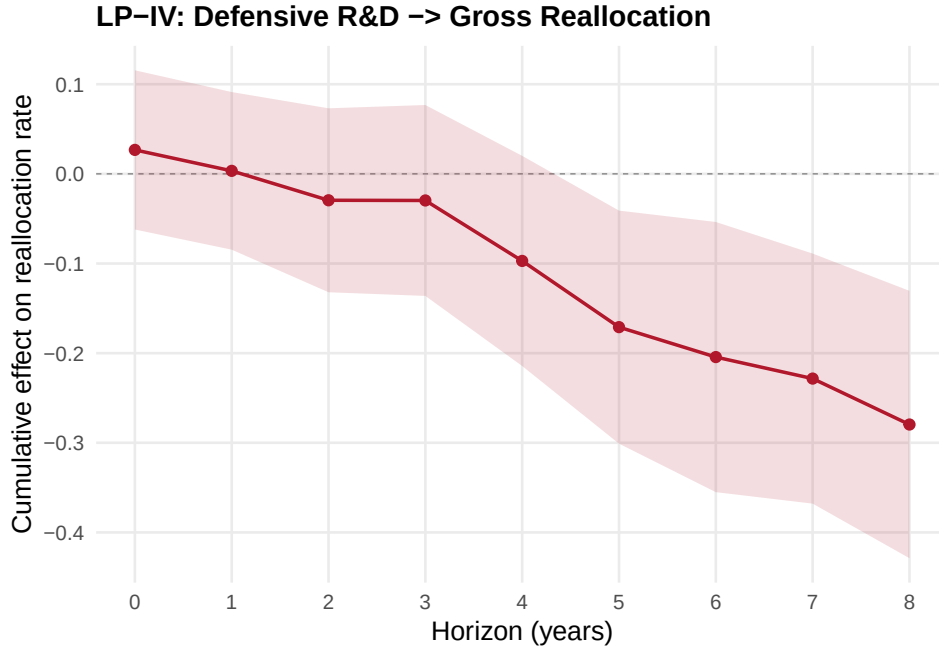


Figure 6: Impulse Response Function of Gross Reallocation

Notes. Gross reallocation = (job creation + job destruction) / employment. Same specification as Figure 5. Shaded region is the 90% CI.

The joint decline in entry and exit is an interesting result. If firms shield their market position or abuse their market power, one would expect exit rates to either stay the same or rise. A possible explanation here is that, with prohibition of new innovation defensive R&D also creates spillovers to the full market, meaning that other firms can profit from a competitors shielding.

4.4 Interpreting the Lag Structure

The common timing pattern across markups, TFP, and dynamism (near-zero at short horizons, significance at horizons 3–5) serves three purposes. First, it discriminates the barrier-accumulation mechanism from alternatives. Reverse causality or common shocks would typically produce contemporaneous effects. Second, the timescale is economically plausible. With a 10% annual displacement probability, it takes several years for reduced displacement to substantially alter the cross-section of surviving firms. Third, the timing matches the patent-to-product pipeline.

4.5 The Mechanism: Revenue versus Cost Productivity

Recall that R&D investments are fixed costs and are therefore expected to raise markups regardless of type (since markups are defined as price over marginal cost). However, the model predicts that productive R&D significantly affects costs, due to the fishing out term in the optimality condition. We should therefore see an increase in both

Table 1: The Markup Channel: Revenue vs. Cost-Based Productivity Growth

	$\Delta_3 \log(\text{Sales}/\text{emp})$	$\Delta_3 \log(\text{COGS}/\text{emp})$
log(Productive R&D)	0.64*** (0.16)	0.013*** (0.006)
log(Defensive R&D)	0.91*** (0.15)	0.01 (0.06)
Industry $\times$ Year FE	Yes	Yes
Observations	15,470	15,470

Notes. Three-year forward growth. Firms with ≥ 3 KPST-scored patents, 1985–2022. Productive R&D = R&D intensity $\times$ breakthrough share. Defensive R&D = R&D intensity $\times$ non-breakthrough share. Standard errors clustered by firm. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

The precision of the zero on cost productivity is notable, and the tight CI can rule out cost-productivity effects larger than one-seventh the revenue-productivity effect.

4.6 Firm Size and the Convexity of Defense

Table 2 tests Prediction 4. The interaction of defensive R&D with log sales is 0.181 ($p < 10^{-6}$): a firm at the 90th percentile of sales experiences a markup effect roughly three times larger than one at the median. The main effect of defensive R&D is indistinguishable from zero ($p = 0.82$), meaning the markup-raising effect is concentrated among larger firms, as equation (7) predicts.

Table 2: Defensive R&D Raises Markups More for Larger Firms

	Markup
Productive R&D	0.475*** (0.118)
Defensive R&D	-0.019 (0.081)
Log(sales)	-0.014* (0.007)
Productive R&D $\times$ Log(sales)	0.195*** (0.045)
Defensive R&D $\times$ Log(sales)	0.181*** (0.035)
Industry $\times$ Year FE	Yes
Observations	19,789
Adj. R^2	0.276

Notes. Markups winsorized at the 1st and 99th percentile. Same sample restrictions as Table 1. Standard errors clustered by firm. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

4.7 Alternative Explanations

Although the findings seem solid in this case, .

Measurement. The KPST non-breakthrough measure captures text-based novelty, not strategic intent. Some non-breakthrough patents may reflect genuine incremental improvements, which would attenuate estimates toward zero. Furthermore, using other proxies for defensiveness yields weak first stage regressions. The coefficients should therefore be interpreted as lower bounds.

General equilibrium. Given the reduced form approach used here, spillover- and GE-effects are not incorporated

Competing mechanisms. The results complement rather than adjudicate between De Ridder (2024) (heterogeneous intangible efficiency), Akcigit and Ates (2023) (declining knowledge diffusion), and Olmstead-Rumsey (2022) (declining laggard quality). These mechanisms are not mutually exclusive and may reinforce each other. Rising barriers reduce the displacement rate, the vehicle through which knowledge spreads in step-by-step models, amplifying the knowledge-diffusion channel. As productive R&D becomes more expensive, laggards disproportionately shift toward incremental innovation, providing an endogenous mechanism for declining laggard quality. Quantifying relative contributions would require structural estimation.

5 Conclusion

If productive R&D becomes progressively harder while defensive R&D does not, firms optimally shift toward defense. I formalized this in a Klette-Kortum model where the

fishing-out asymmetry tilts relative returns toward barrier-building as the frontier advances, endogenously producing rising markups, slowing productivity growth, and declining dynamism. LP-IV estimates show that exogenous increases in defensive R&D cause markups to rise with a 4–6 year lag and TFP to decline from horizon 4 onward. The timing matches the barrier-accumulation mechanism. All six BDS dynamism measures decline uniformly. The revenue-versus-cost decomposition delivers the sharpest test: defensive R&D is associated with increased revenue productivity but not cost-productivity changes.

The distinction between efficient and inefficient barriers carries policy implications. If barriers reflect genuine efficiency then reducing them is undesirable. However, if they reflect defensive R&D, they represent social waste. simple R&D subsidies to increase innovation would not resolve the fundamental incentive problem.

However, several limitations warrant discussion. breakthrough innovation is mostly done by young firms, which are less likely to be publicly listed and hence create sample selection bias. Furthermore, a non-breakthrough patent does not necessarily mean that the innovation was specifically designed to shield the market position., using other proxies, such as the self-citation ratio of a patent, yields first stage F-statistics < 5 , and hence unstable results.

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A Proofs and Derivations.

Growth Equation. Under the Cobb-Douglas aggregator, $\ln Y = \int_0^1 \ln(q_j y_j) dj$. On each product line j , innovators arrive at aggregate rate ι and succeed in displacing the incumbent (who has barrier b_j) with probability $\bar{H}(b_j) = \Pr(\lambda > b_j)$. Conditional on displacement, quality jumps by factor λ , contributing $\ln \lambda$ to $\ln Y$. The growth rate is therefore:

$$g = \int_0^1 \iota \cdot \bar{H}(b_j) \cdot \mathbb{E}[\ln \lambda \mid \lambda > b_j] dj$$

On the BGP the cross-line distribution of barriers is stationary, and writing b^* for the distribution-weighted barrier $\mathbb{E}_\lambda[b(d^*(\lambda))]$, this reduces to $g = \iota \cdot \bar{H}(b^*) \cdot \mathbb{E}[\ln \lambda \mid \lambda > b^*] = \bar{\tau} \cdot \mathbb{E}[\ln \lambda \mid \lambda > b^*]$

Existence of Equilibrium. The firm-size distribution lies on a compact simplex. Innovation and entry rates are bounded by the labor constraint. The composite map from (x, d, e) to the stationary distribution and back to optimal (x, d, e) via the value function is continuous. By Brouwer's fixed point theorem, a balanced growth path exists.

Derivations of Model Predictions **Prediction 1:** The per-product innovation cost is $\ell^P/n = ax^\gamma(1 + \phi Q)^{\gamma-1}$. The marginal cost of x (in research-labor units) is $a\gamma x^{\gamma-1}(1 + \phi Q)^{\gamma-1}$. On the BGP, the FOC equates this to the marginal benefit $\tilde{v}/\tilde{w}_R$ (the value of a new product line, in research-labor units):

$$\tilde{w}_R \cdot a\gamma x^{\gamma-1}(1 + \phi Q)^{\gamma-1} = \tilde{v}$$

Solving:

$$x^* = \left(\frac{\tilde{v}}{\tilde{w}_R a\gamma} \right)^{1/(\gamma-1)} (1 + \phi Q)^{-1}$$

As Q grows, x^* falls at rate $(1 + \phi Q)^{-1}$. The decline in x^* lowers the aggregate flow ι and, through the longer surviving profit stream, raises $\tilde{v}$. By the defense FOC (7), these forces move d^* upward whenever the value effect dominates the flow effect, so d^* is not independent of Q but rises with the frontier. The defensive share of research labor therefore rises. Growth falls because $\bar{\tau}$ depends on x^* through the aggregate innovation flow ι .

Prediction 2: Markups equal quality gaps. When b^* rises, $\bar{\tau}$ falls (fewer displacements) and conditional on displacement, only large steps survive ($\lambda > b^*$). Both effects raise average markups. The lag arises because the markup distribution adjusts gradually as the stock of surviving incumbents changes.

Prediction 3: Entry succeeds with probability $\bar{H}(b^*) = \Pr(\lambda > b^*)$. When b^* rises, $\bar{H}(b^*)$ falls, reducing the expected payoff from entry. Exit rates decline because $\bar{\tau}$ falls.

Prediction 4: Define $F(d, \lambda) \equiv \tilde{v}(\lambda, d) \cdot \iota \cdot h(b(d)) \cdot b'(d) - \tilde{w}_R = 0$, which is (7). By the implicit function theorem,

$$\frac{dd^*}{d\lambda} = -\frac{F_\lambda}{F_d}.$$

$F_\lambda > 0$ because $\partial \tilde{v} / \partial \lambda > 0$. $F_d < 0$ by the second-order condition. Therefore $dd^* / d\lambda > 0$: defense is increasing in product value. Since multi-product firms have higher average λ_j , they choose higher d^* per product and a higher defensive share of total R&D.

B Data and Estimation

Table 3: Summary Statistics

	N	Mean	SD	P25	P50	P75
Panel A. All Compustat firms, 1985–2022						
Sales (\$M)	241,823	2,320	12,826	16.7	113	709
Employees (000s)		9.3	38.5	0.1	0.7	4.8
Total assets (\$M)		3,038	18,972	14.2	133	918
Markup (translog)		1.35	0.79	0.97	1.16	1.47
R&D / Sales		2.79	32.4	0.00	0.00	0.06
Panel B. Patenting firms (≥ 3 KPST-scored patents), 1985–2022						
	23,209					
Sales (\$M)		7,830	24,105	152	952	4,870
Markup (translog)		1.42	0.83	1.01	1.23	1.56
Non-BT share		0.871	0.222	0.833	1.000	1.000
Patents per year		71.9	268.4	4	10	42
R&D / Sales		1.91	22.7	0.026	0.052	0.094
Panel C. Industry-year IV sample (SIC 3-digit), 1985–2021						
	3,315					
Non-BT share		0.902	0.146	0.874	0.961	1.000
Markup (translog)		1.24	0.34	1.01	1.17	1.38
Examiner leniency		0.786	0.111	0.714	0.791	0.862
Patents per ind-yr		495	1,842	12	44	227
Firms per ind-yr		10.8	14.3	3	6	13
Panel D. BDS dynamism (matched industries), 1985–2021						
	2,693					
Entry rate		0.083	0.031	0.062	0.078	0.098
Exit rate		0.085	0.025	0.067	0.082	0.099
Gross reallocation		0.231	0.074	0.179	0.220	0.272
Job creation rate		0.143	0.049	0.108	0.135	0.168
Job destruction rate		0.120	0.041	0.091	0.113	0.142
Firm death rate		0.069	0.028	0.049	0.065	0.084

Notes. Panel A covers all Compustat firms excluding financials (SIC 6000–6999) and utilities (SIC 4900–4999). Panel B restricts to firm-years with ≥ 3 patents with non-missing KPST breakthrough scores. Panel C is the LP-IV estimation sample, consisting of industry-years with ≥ 5 examined patents and non-missing examiner leniency. Panel D adds non-missing BDS dynamism data. Non-BT share is the fraction of KPST-scored patents classified as non-breakthrough. Examiner leniency is the leave-one-out grant rate within art unit $\times$ filing year.

Markups. Following De Ridder et al. (2026), I estimate a translog production function at SIC 2-digit (55 industries with ≥ 200 firm-years): $\log(\text{rev}) = \alpha v + \beta v^2 + \gamma k + \delta + \omega + \varepsilon$, where $v = \log(\text{COGS})$ and $k = \log(\text{PP\&E})$. The output elasticity is $\hat{\alpha}_{it} = \hat{\alpha} + 2\hat{\beta}v_{it}$, and

the markup is $\hat{\mu}_{it} = \hat{\alpha}_{it}/(\text{COGS}_{it}/\text{Rev}_{it})$. Lagged COGS instruments for current COGS. De Ridder et al. (2026) show that revenue-based translog estimates suffer from level bias but capture trends and cross-sectional dispersion well. Since my strategy relies on changes over time and cross-firm comparisons, the level bias is secondary. The resulting markup has mean 1.35, median 1.16, SD 0.79, and correlation 0.83 with the sales/COGS ratio.

TFP. Firm-level TFP is the Solow residual from $\log(\text{VA}_{it}) = \alpha_s \log(K_{it}) + \beta_s \log(L_{it}) + \delta_t + \omega_{it}$, estimated by SIC 2-digit (55 industries with ≥ 200 firm-years). Value added is Sales – COGS + 1; firms with Sales < COGS are dropped. Capital is gross PP&E, labor is employment. Year fixed effects absorb common shocks. TFP is the OLS residual, aggregated to industry-year means for LP-IV estimation (229,975 firm-year estimates). a HP filter with $\lambda = 100$ is used to account for capacity utilization rates on the firm level.

Examiner Leniency IV. I use PatEx 2022 (9.3 million utility applications, 1980–2022). For each art unit $\times$ filing year cell with ≥ 5 examiners, I compute the leave-one-out grant rate. This is averaged across granted patents within each SIC 3-digit $\times$ issue year cell. The identifying assumption is quasi-random assignment within art unit $\times$ year following (Sampat and Williams, 2019).